

Qatar-1b

R-0633

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230901 · created 2026-07-25T17:01:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460188.8695 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.055
Transit depth (Rp/Rs)^2	1.26 +/- 1.24 [%]
Orbital Inclination (inc)	79.18 +/- 2.91
Airmass coefficient 1 (a1)	0.16 +/- 0.11
Airmass coefficient 2 (a2)	1.58 +/- 0.54
Scatter in the residuals of the lightcurve fit is	68.86 %
Best Comparison Star	#3 - [321, 396]
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.018 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.055 vs 0.1463 (deviation 0.55σ)
Duration fit vs published	0.018 d vs 0.0692 d (deviation 1.8σ)
Mid-time O-C	-2.5 min
Depth SNR	1.8
Residual scatter	68.86 %

Light curve

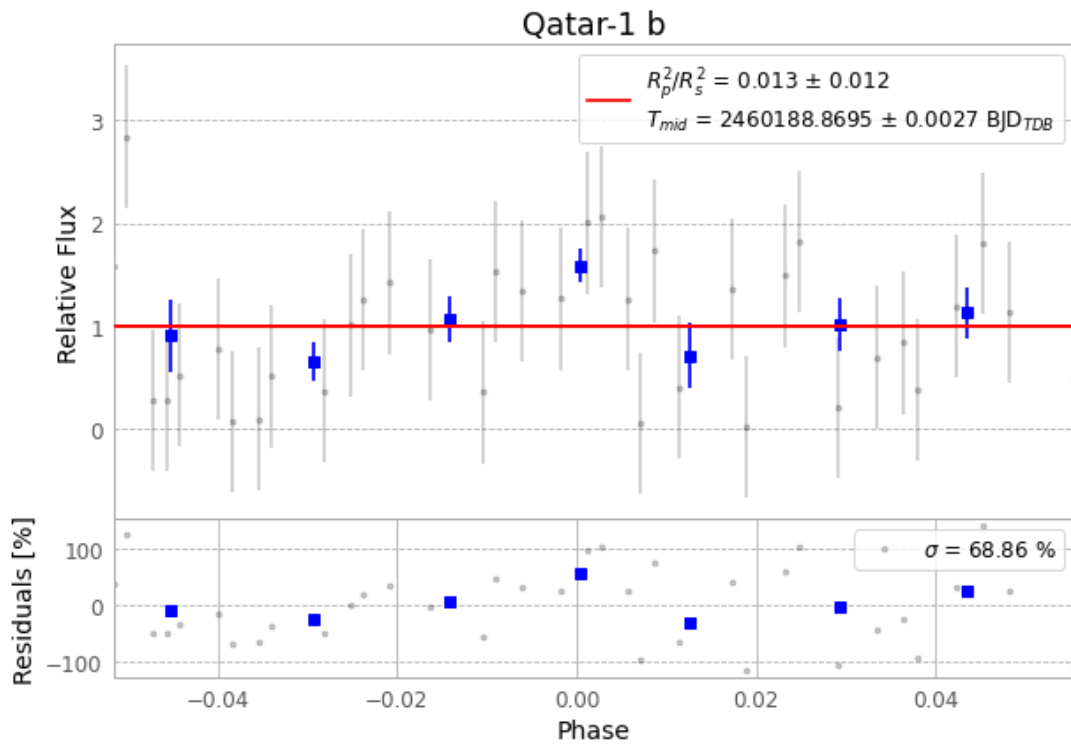


Figure 1 — FinalLightCurve_Qatar-1 b_2023-09-01.png (SHA-256 385adabd76ff90b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 82f98c9369d215c8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1230901070315.FITS → Qatar-1230901104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230901.log (bba4bbcd2b90...), FinalLightCurve_Qatar-1 b_2023-09-01.png (385adabd76ff...), FinalLightCurve_Qatar-1 b_2023-09-01.csv (ae3d49e6ff05...)

Compute resources

Wall time 155.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 17:01Z.